

Comprehensive: area + multiple digits + estimate

Cross-Topic comprehensive application · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5A Units 1 and 2 + Modern Education 5A Unit 1-5

Core traps: □ T1 Estimated unit error + T4 Confusion of large place values + T5 Area formula/unit trap (mixed across topics)

SSPA association: ● **High frequency** Cross-topic comprehensive questions in sub-tests are compulsory every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Class 1-5 (Multi-digit reading and writing and approximate values · Area formula · Unit conversion · Estimation strategy)

The goal of this class: ❶ Estimating the area (estimate the size first → set the formula) ❷ Application of large numbers in area ❸ Comprehensive word questions (reading and writing + estimation + area three-in-one)

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Read the number 2,358,400 and write the place value of "5".	Basic	
2	Estimate: How many meters is the length of a standard basketball court? Circle reasonable answers. A. 2 meters B. 20 meters C. 200 meters	Basic	
3	Calculate the area of the rectangle: length 12 cm, width 8 cm. Area = ?	Basic	
4	Fill in the blank: $5 \text{ m}^2 = \text{_____ cm}^2$ (Hint: $1 \text{ m}^2 = 10000 \text{ cm}^2$)	Basic	
5	Take 3,867,452 to the "hundred thousand" and "ten thousand" places.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Estimate the area - estimate the size first, then apply the formula ● SSPA

- ① **Three-step method for estimating area:** Estimate size → set of formulas → take approximate values
- ② **Step 1 Estimate size:** Observe the shape and use known reference objects to estimate the length (for example: one step is about 1m, one floor is about 3m)
- ③ **Step 2 set of formulas:** Rectangular area = length $\times$ width; square area = side length $\times$ Side length
- ④ **Step 3 Take the approximate value:** Round the calculation result to the appropriate place value (thousands/ten thousand), and mark "approximately" when answering
- ⑤ **Trap T1:** The units must be consistent when estimating - you cannot directly multiply the length by m and the width by cm!



Figure 1: Estimation of football field area - first estimate the length and width, and then use the area formula

□ Trap detonation example (cross-topic mixed trap T1+T5)

Example 1: The picture below is the floor plan of a rectangular park. Estimate its area and express it in "10,000 m²".

✗ **Common mistakes**

✓ **Correct solution**

Length=200m Width=150m

$$\rightarrow 200+150=350$$

Use the perimeter formula (addition) to calculate the area (should be multiplied) or: forget to take the approximate value/wrongly write the unit

$$200 \times 150 = 30,000 \text{ m}^2$$

$$= 30,000 \text{ m}^2$$

Step1: Length $\approx$ 200m Width $\approx$ 150mStep2:
Area=length $\times$ width=30,000m²Step3: 30,000=30,000

example

Example 2: A square square with side length about 85 m. Estimate its area and round your answer to the nearest thousand.

⚠ T1 trap: The units for estimating length and width must be the same (both in m or both in cm) and cannot be mixed!

⚠ T5 trap: The area formula is "multiplication" not "addition"! The area of a rectangle = length $\times$ width, not length + width.

Knowledge Point 2: Application of Large Numbers in Area - Conversion of m² and cm² ● SSPA Required Exam

- ① **Key conversion:** 1 m = 100 cm $\rightarrow$ **1 m² = 100 $\times$ 100 = 10,000 cm²**(not 100!)
- ② m² $\rightarrow$ cm² : $\times$ **10,000**(the number will become four digits larger)
- ③ cm² $\rightarrow$ m² : $\div$ **10,000**(the number will become four digits smaller)
- ④ m² or km² is commonly used for large areas (parks, cities); cm² is commonly used for small areas (books, desktops)
- ⑤ **Trap T4:**After conversion, the numbers are very large (millions, tens of millions), and it is easy to get the bit value wrong when reading and writing.

①

Remember the conversion rates

$$1 \text{ m}^2 = 10,000 \text{ cm}^2$$

②

m² $\rightarrow$ cm²

$\times$ 10,000 numbers become larger

③

cm² $\rightarrow$ m²

$\div$ 10,000The number becomes smaller

④

Check plausibility

The number in cm² must be larger than m²

example

Example 3: The area of a playground is 3,500,000 cm². How many m² is it equal to? (write down the calculation process)

example

Example 4: The area of a house is 85 m². Equal to how many cm²? Read the numerical value of the answer.

Knowledge point 3: Comprehensive application questions - multi-digit reading and writing + estimation + area calculation 🟡 SSPA killer questions

- ① **Common comprehensive question types for sub-tests:** Test readings simultaneously in one question → Take approximate values → Calculate area with approximate values → Convert units
- ② **Problem-solving strategy:** Break down step by step, each sub-question is processed independently, the answer of the previous step may be the input of the next step
- ③ **Trap T4+T5 Mixed:** The numbers change after taking approximations, and the area formula is easy to make mistakes; converting units involves multiplication and division of large numbers
- ④ **Must check:** Is the unit of each answer correct? Are approximations marked "approximately"? Is the reading of large numbers correct?

📌 Comprehensive trap example questions (T1+T4+T5 three-in-one)

Example 5: A rectangular square is about 125 m long and 85 m wide.

- (a) Estimate the area of the square to the nearest thousand.
- (b) Read the answer to (a).
- (c) Convert the answer to (a) into cm^2 .

example

Example 6: Round 2,468,000 to the 100,000th place to get A. Using A as the side length of the square (unit: m), calculate the area of the square and express it in " $10,000 \text{ m}^2$ ".

🧠 **Tip: "When estimating, first estimate the length and width. Multiply the area instead of adding. m^2 becomes cm^2 multiplied by 10,000. Check the reading and writing of large numbers."**

⚠️ **Comprehensive questions are the most common mistakes: after taking the approximate value and forgetting to mark the word "approximately", one point will be deducted from the test! If the answer is calculated using approximate values, write " $\approx$ " throughout.**

III. Lesson Layered Synchronization Practice

Basic layer (total 6 questions, everyone must do)

#	Question	Difficulty	Working Space
6	Estimate the area of a rectangular playground: about 30 m long and 20 m wide. Area approx = ?	🌱	
7	$4 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$	🌱	
8	A rectangle is 15 m long and 12 m wide. Area = ? m^2	🌱	

9	Read the number 3,050,700 and write down the values of the two "0"s in it.		
10	Take 4,567,890 to the "100,000 digit".		
11	6 m ² = _____ cm ² . Read the answer.		

Advanced layer (total 5 questions, choose do)

#	Question	Difficulty	Working Space
12	A park is approximately 250 m long and 160 m wide. Estimated area, expressed in "10,000 m ² ".		
13	The area of a playground is 4,800,000 cm ² . How many m ² is it equal to?		
14	Compare two sites: Site A (95 m long and 65 m wide) and Site B (85 m long and 75 m wide). Which one has the larger area? How many m ² is the difference?		
15	The area of a city park is 25,000,000 m ² . (a) Read this number (b) Express it in "square kilometers" (1 km ² = 1,000,000 m ²)		
16	Round 4,567,890 to the "hundredthousandth digit" and use this approximate value as the side length of the square (unit: m) to estimate the area of the square.		

challenge layer (total 5 questions, choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
17	A parking lot is approximately 135 m long and 85 m wide. (a) Estimate the area to the nearest thousand (b) If each car requires 25 m ² , approximately how many cars can be parked at most?		
18	A rectangular site is 125 m long and 80 m wide. (a) Area = ? m ² (b) Round the answer to the nearest ten thousand (c) Convert the original area to cm ²		
19	Comprehensive question: First, round 3,856,200 to the 100,000th place to get A, and then use A as the length of the rectangle (width = 100 m). Estimate the area of this rectangle and read the answer.		
20	A square park with sides approximately 200 m. (a) Estimate the area to the nearest hundred thousand (b) If 4 trees are planted in 1 m ² , approximately how many trees can be planted? (c) Round the number of trees to the 10,000s		
21	A square is approximately 180 m long and 120 m wide. (a) Estimate the area (b) Round the answer to the nearest tens of thousands (c) If the cost of the square is \$250 per m ² , estimate the total cost (round the answer to the nearest million)		

Knowledge point 4: Comprehensive application questions (subject to sub-test text questions)

 **SSPA compulsory exam**

Four steps to solve the problem: ① **Circle the key numbers and units**(pay attention to m or cm) ② **Judgment operation**(Estimation? Area? Conversion?) ③ **Calculate step by step**(Write each step clearly) ④ **Write a complete answer**+ Inspection unit

comprehensive application question (total 8 questions, all must do)

#	Question	Difficulty	Working Space
22	The school hall is 35 m long and 22 m wide. If tiles are to be laid, 9 tiles are needed per m^2 . How many tiles are needed in total? (Calculate the area first)		
23	A swimming pool is 50 m long and 25 m wide. There is a 2 m wide walkway around the pool. What is the area of the sidewalk in m^2 ? (Tip: large rectangle – swimming pool)		
24	A theme park in Hong Kong covers an area of approximately 1,260,000 m^2 . (a) Write how this number is read (b) Round it to the nearest hundred thousand (c) If there are 50,000 visitors per day, use an approximation to estimate the average m^2 available per person? (Answer is rounded to an integer)		
25	A rectangular farmland is 320 m long and 180 m wide. Express the area in "hectares" (ha) (1 ha = 10,000 m^2), and round the answer to one decimal place.		
26	Express the area of a rectangular garden (8 m long and 6 m wide) in m^2 and cm^2 respectively. Comparing two numbers, why is the number cm^2 so much larger?		
27	A large shopping mall has an area of 850,000,000 cm^2 . (a) How many m^2 is it equal to? (b) If a shopping mall has 5 floors and each floor has the same area, what is the area in m^2 of each floor?		
28	The areas of the three parking lots are: A 1,250,000 m^2 , B 980,000 m^2 , C 1,580,000 m^2 . (a) Round each to the 100,000th place (b) Use approximations to find the maximum and minimum area difference (c) Read the answer to (b)		
29	A rectangular site is approximately 178 m long and 122 m wide. (a) The estimated area is rounded to the nearest ten thousand (b) If one building is built every 100 m^2 , approximately how many buildings can be built? (calculated using approximate values)		

IV. Ultimate challenge area (total 3 questions, choose do, SSPA finale level)

#	Question	Difficulty	Working Space
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 1	A city park covers an area of approximately 3,800,000 m ² . (a) Read this number (b) Convert to km ² (1 km ² = 1,000,000 m ²) (c) If the park receives visitors every day, and each person uses an average of 20 m ² of space, how many people can it receive at most per day? (rounded to the nearest ten thousand)		
 2	Round 2,468,135 to the 100,000th place to get A, and round to the 10,000th place to get B. (a) How different are A and B? (b) Use A as the side length of the square (unit: m), area = ? m ² (c) Use B as the length of the rectangle (width = 50 m), area = ? m ²		
 3	A rectangular site approximately 156 m long and 98 m wide. (a) Estimate the area to the nearest thousand (b) Using the approximation of (a), if one stall is provided for every 25 m ² of the venue, approximately how many stalls can be installed? (c) If the rent for each stall is \$380, what is the approximate total rent? (Take to the nearest 100,000 digits)		

V. Class afterhomework

Basic must-doquestions (total 5 questions, mustwrite outcalculateprocess)

#	Question	Difficulty	Working Space
H1	Estimate: A rectangular court approximately 40 m long and 25 m wide. Area approx = ? m ²		
H2	8 m ² = _____ cm ² . Read the answer.		
H3	Read 2,500,000 and write the place value of "2".		
H4	A playground is 60 m long and 35 m wide. (a) Area = ? m ² (b) Convert to cm ²		
H5	Take 5,678,900 to the "100,000" and "10,000" digits.		

Advanced choose doquestion (total 3 questions, choose do)

#	Question	Difficulty	Working Space
H6	A park is approximately 150 m long and 120 m wide. The area is estimated and expressed in "10,000 m ² ".		
H7	The rectangular square is 85 m long and 65 m wide. (a) Area = ? m ² (b) If 4 bricks are laid per m ² , how many bricks are needed? (c) Count the number of bricks to the 10,000s		
H8	Comparison: Site A has an area of 2,400,000 cm ² and Site B has an area of 250 m ² , which one is larger? How many m ² is the difference? (Tip: Unify the units first)		

VI. The Lessoncorecommon errorssummary

✅ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🧩 basic questions independently

☐ I can challenge 🌿 advanced questions

☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve 🧩 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	$m^2 \leftrightarrow cm^2$ Multiplication error SEP : $1 m^2 = 100 cm^2$ (wrong!): $1 m^2 = 100 cm^2$ (wrong!)	$1 m = 100 cm \rightarrow 1 m^2 = 100 \times 100 = 10,000 cm^2$
2	I forgot to write "approximately" and the unit when making an estimate.	The estimated answer must be written as "about" + unit (m^2/cm^2)
3	Bit value confusion when reading and writing large numbers SEP : one hundred thousand vs ten thousand vs one million: One hundred thousand vs ten thousand vs one million	From the right: one hundred thousand/ten thousand hundred thousand million. group of four
4	After taking the approximate value, use the approximate number to calculate, and forget to mark "$\approx$"	Each step of calculation using approximate values must be marked " $\approx$ ", and no points will be deducted when submitting the test.
5	Wrong use of area formula SEP : Use "length + width" instead of "length $\times$ width": Replace "length $\times$ width" with "length + width"	Area = length $\times$ width (multiply, not add!). Perimeter is plus
6	Confusing different units in comprehensive questions SEP : mixed operations of m^2 and cm^2: m^2 and cm^2 mixed operations	Unify the units before calculating! Convert all to m^2 or all to cm^2
7	Missed sentences/missed units in application questions	Must write "Answer:..." + unit, otherwise step points will be deducted

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📖 Related topics: L03 Area of parallelograms and triangles • L04 Area of trapezoidal polygons • L05 Special area trap